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Lunar polymict breccia 14321: a petrographic study

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Abstract—Seven petrographic thin sections of lunar rock sample 14321, 'Big Bertha', have been examined. It is a complex rock incorporating diverse lithic and single crystal fragments and represents a sampling of the heterogeneous Fra Mauro formation, considered by the writers to be lithified debris from the Imbrium impact event. Electron probe microanalysis and microscopic study of textures reveal the assembly history of this breccia which in turn allows some interpretation of the nature of the pre-Imbrium crust and the effect of the Imbrium impact and the subsequent transportation to the Apollo 14 site. The present-day polymict breccia 14321 is composed of basaltic clasts originating from the fragmentation of a single or closely related set of lava cooling units, a set of fragmental clasts designated as microbreccia 3 (themselves polymict microbreccias), and a light colored matrix which formed rock 14321 by cementing the two major groups of clasts. The light colored matrix material is derived from the fragmentation and mutual abrasion of the basalt and microbreccia 3. On the basis of consistent textural relations two older sets of microbreccias have been identified within microbreccia 3. Microbreccia 1 clasts are well-rounded, relatively light colored, and noritic. They are always completely enclosed within microbreccia 3, most often forming the central cores of rounded accretionary lapilli structures which we have designated as microbreccia 2. Microbreccias 1, 2, 3, and macrobreccia 14321 represent a chronological series of fragmentation and lithification events. Each of these events involved some thermal and/or shock metamorphism as evidenced by mineralogical and textural criteria, and the chronological order of formation of the breccias also corresponds to a decreasing intensity of associated thermal effects. The petrology and mineralogy of 14321 are described in detail in this paper. A more general interpretation of the combined petrographic and chemical data is given in DUNCAN *et al.* (1975a).

INTRODUCTION

ONE OF the primary scientific objectives of the Apollo 14 mission was to sample the Fra Mauro formation which is widely distributed in a broad belt around Mare Imbrium. The Fra Mauro formation is believed to be composed of material excavated by the large impact explosion which formed the circular Imbrium basin. The initial Imbrium ejecta blanket material has been reworked to varying degrees by subsequent smaller impact events. In the immediate area of the Apollo 14 landing site, Cone Crater is sufficiently large (approximately 340 m diameter) to have penetrated relatively undisturbed Fra Mauro material beneath the local regolith. Consequently, large blocks of Cone Crater ejecta could contain old (pre-Imbrium event) fragments which still retain some of the mineralogical, textural, and chemical features characteristic of an ancient, near-surface Moon layer as well as some imprints of the fragmentation, transportation, and deposition processes related to the Imbrium impact.

Sample 14321 was picked up near the edge of Cone Crater, and, at the time of the Apollo 14 mission, it was the largest (9·0 kg) coherent rock sample returned to

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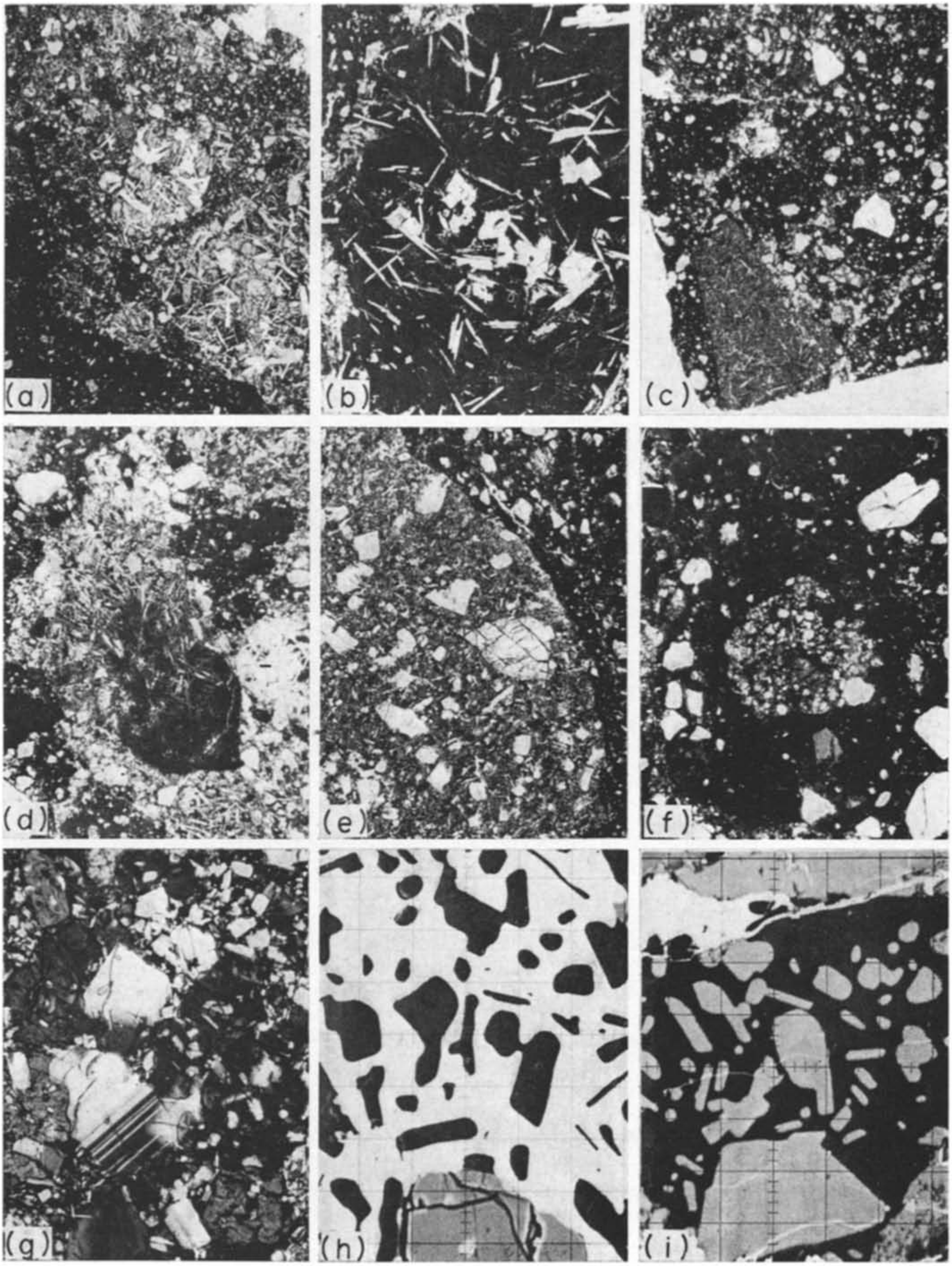

Fig. 1. (a–i)

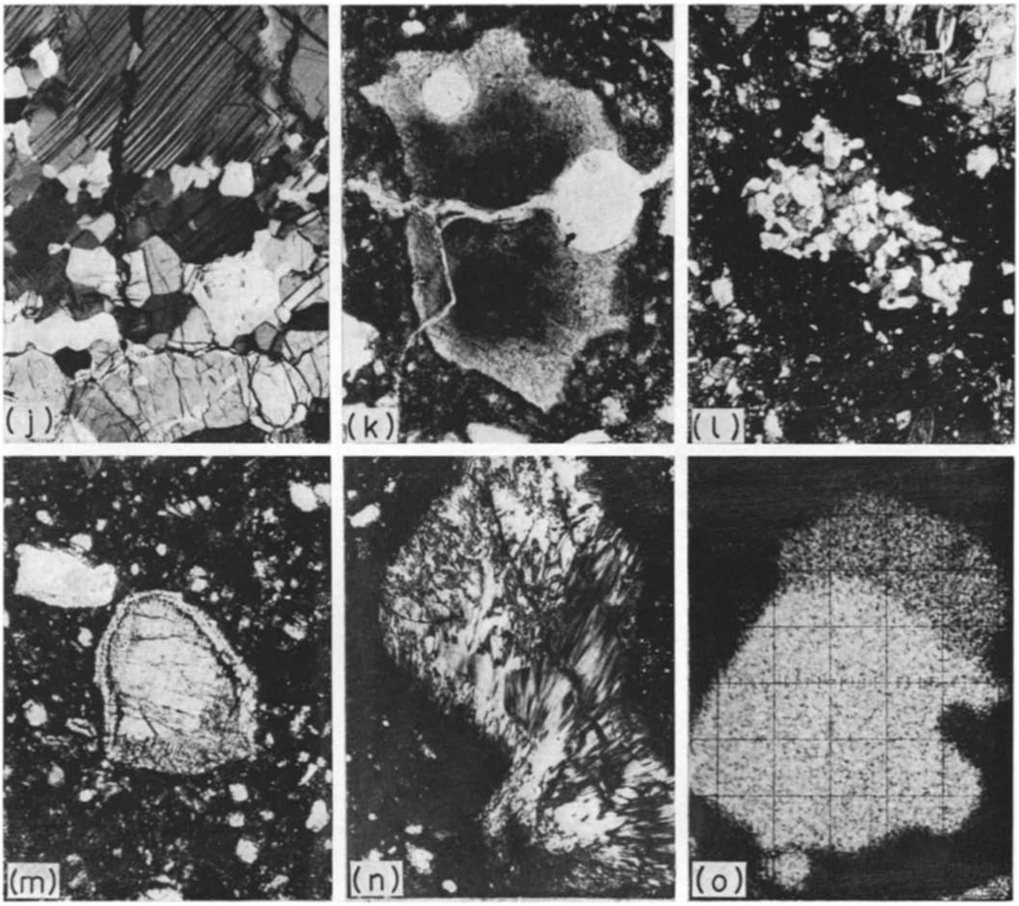

Fig. 1. Photomicrographs of 14321. Distances in parentheses correspond to width of the photograph. (a) Two clasts of ophitic basalt surrounded by light matrix. Lower left of photo is dark matrix of microbreccia 3 (6 mm). (b) Vitrophyric basalt clast surrounded by light matrix. Dark portion is glass (1·5 mm). (c) Portion of elongate variolitic basalt clast at lower left. Several rounded microbreccia 2–3 clasts can also be seen. All clasts are separated by thin seams of light matrix (6 mm). (d) Glassy basalt clast showing incipient variolitic crystal texture. Microbreccia 2–3 clasts and light matrix also visible (3·8 mm). (e) Portion of microbreccia 1 (noritic) clast. Note that clast is well-rounded and that matrix is coarser and lighter colored than matrix of surrounding microbreccia 3 (1·5 mm). (f) Microbreccia 2 (lapilli structure?) with a microbreccia 1 core (1·5 mm). (g) Micronorite texture. Note poikiloblastic pyroxenes. Texture suggests thermal metamorphism of igneous norite in contrast with obvious fragmental origin of microbreccia 1 in (e) (0·6 mm). (h) Zircon (white)–plagioclase (black) intergrowth in micronorite. BSE image (115 μm). (i) Whitlockite (black)–plagioclase (gray) intergrowth in micronorite. BSE image (170 μm). (j) Portion of plagioclase-rich clast in microbreccia 3. Note lamellae offset by microfault and mosaic recrystallization of plagioclase. Subordinate amount of anhedral olivine is also present at bottom (960 μm). (k) Glassy (devitrified) rhyolite clast with vesicles (0·6 mm). (l) Microgranite clast forming core of microbreccia 2 (2·4 mm). (m) Ca-rich pyroxene in microbreccia 3. Note exsolution lamellae of orthopyroxene and reaction rim of pigeonite (0·6 mm). (n) Devitrified glass fragment of plagioclase composition (0·6 mm). (o) Composite metal grain in microbreccia 3. Ni X-ray scan (28 μm).

Earth. In common with many lunar breccias it is polymict, i.e. it contains fragments of pre-existing lithic units, and the different types of clasts represent a series of brecciation and lithification events which may be put into a relative time sequence. Lunar fragmental rocks characteristically afford a rich variety of lithic types and mineralogy for petrographic and chemical analysis. This characteristic is regarded as a mixed blessing by many lunar sample investigators (ourselves included) because the interpretation of the wealth of analytical data can often boil down to nothing more than an ode to the rock's complexity. Nevertheless, we felt that a major effort at petrographic analysis of 14321 was desirable for several reasons. Its size, the nature of its geologic setting, and the now considerable background of plausible ideas about early lunar petrologic processes made it more likely that the petrographic data might be interpreted in a wider context than the confines of a polished thin section or a composition triangle.

A detailed petrographic description of an isolated rock sample, even a lunar sample, must be justified mainly as a necessary prelude the to intelligent interpretation of petrogenesis and more general geochemical and geophysical data. We believe that the interpretations of the petrographic data set forth in this paper are valid within a wider context than the confines of seven thin sections, and the more general conclusions are presented in a companion paper (DUNCAN *et al.*, 1975a).

PETROGRAPHY

1. *General description*

Many lithic units can be recognized in breccia 14321 on a macroscopic scale (see Fig. 1 in DUNCAN *et al.*, 1975b). Based upon a more detailed look at the thin sections we have found it possible to organize these into three broad genetic classes: (1) rounded fragments of pre-existing microbreccia with microscopically recognizable crystal and lithic fragments incorporated in a dark matrix, (2) igneous rock fragments of basaltic composition, and (3) a light colored, somewhat friable matrix which holds 14321 together. Although this broad classification scheme hides much complexity within the individual groups, it does emphasize what we believe to be the principal building blocks of 14321. The following petrographic description is based on this three-fold division and will introduce the complexities as each group is described in detail.

2. *Basaltic fragments and light matrix*

The basaltic fragments form a series which includes glassy, vitrophyric, variolitic, and fine to medium grained ophitic textures (Figs. 1a–d). Most of the fragments are sub-rounded, medium grained ophitic basalts. Evidence of shock metamorphism is limited to fractures which are concentrated at the outer edges of the clasts. The holocrystalline fragments show little mineralogical variation other than the absence or presence of olivine. Microprobe analyses of seven glassy and fine grained (variolitic) clasts are given in Table 1, no. 1. Small portions ($\sim$20 mg) of four of the coarser grained basaltic clasts were mined from bulk sample 14321,184 and fused at 1400°C and $f_{O_2} = 10^{-12}$ atm. Microprobe analyses of the corresponding quenched glasses are given in Table 1, no. 2. The chemical compositions of the glassy and fine grained basalts show very little variation in spite of the obvious textural differences. Due to the small (20 mg) samples, the glasses formed from the coarser grained clasts display a predictably higher dispersion about the average major element concentrations. The differences in average composition between the holocrystalline and partly glassy basalt clasts are most easily explained in terms of the unavoidable sampling error in the small samples of the former. There is no reason to suppose that all the basaltic clasts did not originate from the fragmentation of a single lava cooling unit or a genetically related series of flows. The textural variety is most probably related to variations in cooling rates due to position in the cooling unit(s).

The basaltic fragments are generally separated from the dark breccia clasts by narrow seams of light matrix material (Figs. 1a, c, d). The light matrix mineralogy resolvable under the petrographic microscope is generally similar to that of the basalts, and in places the light matrix includes small fragments of ophitic texture characteristic of most of the basalt fragments. We interpret the light matrix

Table 1. Defocused beam microprobe analyses of various lithologies in 14321

	1	s.d.	2	s.d.	3	s.d.	4	s.d.	5	6	7	s.d.	8	s.d.	9
SiO_2	46·82	0·54	48·06	1·86	46·01	1·25	46·50	0·47	49·56	43·36	72·17	1·96	74·19	1·55	78·73
TiO_2	2·33	0·19	2·42	0·14	2·08	0·16	0·59	0·17	0·24	0·01	0·58	—	0·32	0·14	—
Al_2O_3	13·81	0·43	11·91	1·14	17·49	1·23	21·48	1·68	23·69	29·23	12·00	0·73	12·24	0·73	10·80
Cr_2O_3	0·32	0·03	0·21	0·02	0·13	0·02	0·04	0·01	0·09	0·00	0·16	—	0·09	0·07	—
FeO	15·92	0·31	15·31	3·45	10·07	1·04	6·03	0·51	4·52	4·49	1·85	0·45	1·47	0·62	0·26
MnO	0·26	0·04	—	—	0·10	0·03	0·15	0·02	0·06	0·11	0·06	—	0·04	0·02	—
MgO	6·30	0·29	9·42	2·24	9·18	1·10	6·00	1·18	5·23	5·45	0·09	0·03	0·17	0·08	0·12
CaO	11·57	0·16	10·87	0·91	10·01	0·68	12·67	0·70	12·51	15·90	1·57	0·33	1·32	0·23	0·63
BaO	0·15	0·11	—	—	0·24	0·08	0·31	0·03	0·34	0·28	1·56	0·50	0·60	0·18	0·54
Na_2O	0·80	0·10	0·64	0·11	0·96	0·13	0·83	0·12	0·84	0·59	0·23	0·09	0·30	0·14	0·24
K_2O	0·25	0·08	0·18	0·07	0·73	0·17	0·56	0·46	1·30	0·14	8·27	0·38	8·85	0·44	9·17
P_2O_5	0·25	0·02	—	—	0·97	0·22	0·43	0·33	0·13	0·18	—	—	0·03	0·03	—
Total:	98·78		99·02		97·97		95·59		98·51	99·74	98·54		99·62		100·49
Molecular norms*															
$FeCr_2O_4$	0·4		0·2		0·1		0·1		0·1	0·0	0·2		0·1		—
$Ca_5P_3O_{9\cdot5}$	0·6		—		2·1		0·9		0·3	0·4	—		0·1		—
$FeTiO_3$	3·4		3·5		3·0		0·9		0·3	0·0	0·9		0·5		—
$KAlSi_3O_8$	1·6		1·1		4·4		3·5		7·8	0·8	51·8		54·3		55·8
$NaAlSi_3O_8$	7·5		6·0		8·9		7·9		7·7	4·5	2·2		2·8		2·2
$BaAl_2Si_2O_8$	0·3		—		0·4		0·6		0·5	0·5	3·0		1·1		1·0
$CaAl_2Si_2O_8$	34·8		30·2		42·0		55·3		57·5	75·9	4·7		5·0		0·4
$CaSiO_3$	9·6		10·3		1·0		3·1		1·9	0·6	1·4		0·7		1·1
$MgSiO_3$	18·3		21·0		22·6		17·4		14·7	0·4	0·3		0·5		0·3
$FeSiO_3$	22·3		27·0		11·4		8·9		6·6	0·2	1·9		1·7		0·4
$MnSiO_3$	0·4		—		0·2		0·2		0·1	0·2	0·1		0·1		—
Mg_2SiO_4	0·0		0·0		2·6		0·0		0·0	10·9	0·0		0·0		0·0
Fe_2SiO_4	0·0		0·0		1·3		0·0		0·0	5·1	0·0		0·0		0·0
SiO_2	0·8		0·7		0·0		1·2		2·5	0·0	33·5		33·1		38·8

1. Av. of 7 vitrophyric and variolitic basalt clasts. 2. Av. of fused beads from 4 holocrystalline basalt clasts. 3. Av. of 14 areas dark matrix of microbreccias 2 and 3. 4. Av. of 3 microbreccia 1 clasts. 5. Av. of 2 micronorite clasts. 6. Feldspathic lithic clast. Norm contains 0·5 Ne. 7. Av. 6 high Si areas, mesostasis, basalt clasts. 8. Av. of 9 rhyolite glass clasts. 9. Granoblastic K-microgranite clast.

* Σ cations = 1 for each molecule.

material as being largely made up of finely fragmented basalt, very similar, if not identical, to that found as distinct clasts in 14321. Accordingly, the mineralogical data for both components are presented together.

Subhedral grains of olivine up to 600 μm diameter are present in some of the basalts. In thin section the grains usually show evidence of magmatic resorption. Individual olivine grains are normally zoned with respect to Fe/Mg. The total range of olivine composition found in the basalt fragments is Fa_{24-46} (65 analyses). The Cr_2O_3 content is $0{\cdot}08 \pm 0{\cdot}02$ wt.% (average of 55 analyses, $\pm 1\sigma$) and shows no correlation with Fe/Mg, in contrast with olivines in Apollo 11 and 12 basalts which are richer in Cr_2O_3 and show a strong correlation of Cr_2O_3 with Fe/Mg (Wood *et al.*, 1971). Numerous inclusions of Ti-chromite in the olivines indicate that much of the Cr_2O_3 had been partitioned from the liquid at the time of olivine crystallization. Olivines with very similar compositional range, normal zoning, and low Cr_2O_3 content are also found in the light matrix. These grains are usually subrounded and fractured. A complete analysis of an olivine in an ophitic basalt clast is given in Table 2, no. 1.

Strongly zoned pigeonite and augite are common in the basalt fragments and in the light matrix, but orthopyroxenes are absent. Many of the clinopyroxene

Table 2. Composition of minerals

	1	2	3	4	5	6	7	8
SiO_2	36·49	52·94	49·80	44·57	0·00	—	98·17	51·61
TiO_2	0·06	0·46	1·37	—	53·91	3·26	—	1·37
Al_2O_3	0·09	1·07	1·94	34·92	0·12	18·15	0·81	2·74
Cr_2O_3	0·06	0·71	0·61	—	0·38	43·17	—	0·58
FeO	30·88	17·21	16·08	0·48	41·92	29·79	0·12	4·04
MnO	0·32	0·32	0·33	0·00	0·38	1·31	—	0·11
MgO	31·64	23·18	15·59	0·20	3·37	5·30	0·02	16·65
CaO	0·27	3·05	13·05	19·45	—	—	0·13	22·86
Na_2O	—	0·02	—	0·49	—	—	—	0·04
K_2O	—	—	—	0·02	—	—	—	0·00
Total:	99·81	98·96	98·77	100·13	100·08	100·98	99·25	100·00
O≡	4	3	3	8	3	4	2	3
Si	0·994	0·981	0·953	2·061	—	—	0·991	0·946
Ti	0·001	0·007	0·020	—	0·996	0·080	—	0·019
Al	0·003	0·023	0·044	1·901	0·003	0·698	0·010	0·059
Cr	0·001	0·010	0·009	—	0·007	1·114	—	0·008
Fe	0·703	0·267	0·257	0·019	0·861	0·786	0·001	0·062
Mn	0·008	0·005	0·005	0·000	0·008	0·036	—	0·002
Mg	1·285	0·641	0·445	0·014	0·124	0·258	0·000	0·455
Ca	0·008	0·061	0·268	0·965	—	—	0·001	0·449
Na	—	0·000	—	0·044	—	—	—	0·001
K	—	—	—	0·001	—	—	—	0·000

1. Olivine, basalt clast. 2. Pigeonite core of pyroxene, basalt clast. 3. Augitic pyroxene Ti-chromite spinel, inclusion in olivine of basalt clast. 7. Cristobalite, basalt clast. 8. Diopside crystal clast in microbreccia. 11. Pigeonite rim on 10, crystal clast in microbreccia. 12. Plagio- 14. Orthopyroxene, micronorite lithic clast. 15. K-feldspar, K-microgranite lithic clast. Total

grains are composite with pigeonite cores rimmed by augite. A typical example of the total range in composition and the general trend of the gradients found in grains from a single basalt clast is shown in Fig. 2a. The compositional break between pigeonite core and augite rim is usually sharp. Small, interstitial pyroxene grains are also plotted in Fig. 2a and show general Fe enrichment in addition to grain to grain variability. Rapid crystallization from an Fe-rich liquid is indicated for these late pyroxenes. The composition range for pyroxene from all the basalt clasts as well as the light matrix is almost identical to that shown in Fig. 2a. Representative total analyses of pigeonite and augite are given in Table 2, nos. 2 and 3.

Plagioclase is abundant in the basaltic clasts and the light matrix. The total compositional range found in all basalt clasts is $An_{96}Ab_4Or_0$ to $An_{72}Ab_{24}Or_4$. Normal zoning is pronounced in the medium grained basalts, and the range within some individual crystals is close to that given above. A frequency distribution of An concentration for random spot analyses of plagioclase from the basalts is shown in Fig. 3. A similar plot for plagioclase from the light matrix is also shown for comparison. Both plots are based on a minimum of 50 individual analyzed grains, and the close correspondence serves to reinforce the idea of a similar provenance. A complete analysis of plagioclase from a sub-ophitic basalt clast is given in Table 2,

in various lithologies of 14321

9	10	11	12	13	14	15	
54·89	51·59	54·47	44·37	0·13	53·80	63·33	SiO_2
0·52	1·68	0·90	0·03	0·31	0·50	—	TiO_2
1·31	2·92	0·83	36·02	56·56	0·63	18·36	Al_2O_3
0·36	0·51	0·36	—	9·44	0·33	—	Cr_2O_3
10·45	10·09	15·26	0·10	12·82	22·14	0·11	FeO
0·16	0·20	0·27	—	0·17	0·34	—	MnO
30·85	15·45	24·17	0·06	19·11	20·79	0·00	MgO
1·23	17·62	3·69	19·48	0·00	1·79	0·31	CaO
0·00	0·14	0·03	0·31	0·00	0·01	0·29	Na_2O
0·00	0·03	0·02	0·03	0·00	0·01	15·11	K_2O
99·77	100·23	100·00	100·40	98·54	100·34	99·40	Total
3	3	3	8	4	3	8	≡O
0·972	0·954	0·989	2·042	0·004	0·998	2·977	Si
0·007	0·023	0·012	0·001	0·006	0·007	—	Ti
0·027	0·064	0·018	1·954	1·762	0·014	1·017	Al
0·005	0·008	0·005	—	0·197	0·005	—	Cr
0·155	0·156	0·232	0·004	0·283	0·344	0·004	Fe
0·002	0·003	0·004	—	0·003	0·005	—	Mn
0·814	0·426	0·654	0·004	0·753	0·575	0·000	Mg
0·023	0·349	0·072	0·961	0·000	0·036	0·015	Ca
0·000	0·006	0·001	0·028	0·000	0·000	0·027	Na
0·000	0·001	0·000	0·002	0·000	0·000	0·906	K

rim on 2. 4. Plagioclase, basalt clast. 5. Ilmenite, inclusion in pyroxene, basalt clast. 6. host, crystal clast in microbreccia. 9. Bronzite exsolution lamella in 8. 10. Augite core, clase crystal clast in microbreccia. 13. Cr-pleonaste spinel, crystal clast in microbreccia. includes 1·89 wt.% BaO.

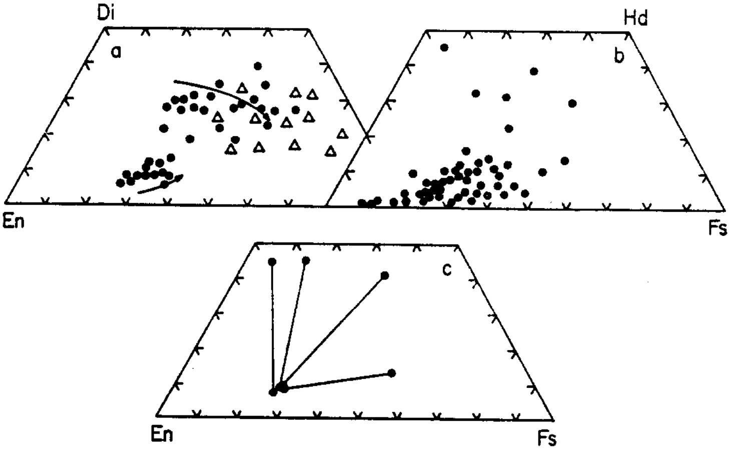

Fig. 2. (a) Compositional trends in clinopyroxene from ophitic basalt clast. Generalized zoning in individual pigeonite–augite composite grains shown in arrows. Triangles are spot analyses of smaller, interstitial crystals. (b) Random spot analyses of crystal fragments in microbreccia 2–3 clasts. Large majority of the grains are orthopyroxenes or pigeonites. (c) Representative range of compositions for pigeonite reaction rims and core pyroxene clasts in microbreccia 2–3.

no. 4. The atomic proportions based on 8 oxygen atoms indicate that (Si–Na–K)-2 = 0·016 and 1-(Al–Ca–Fe–Mg) = 0·097. These departures from ideal feldspar stoichiometry fall on the general trend that has been detected for other lunar basalts (DRAKE and WEILL, 1971).

Ilmenite is the most abundant opaque mineral in the basalts and light matrix. Its textural associations in the basalt clasts range from inclusions in augitic pyroxene

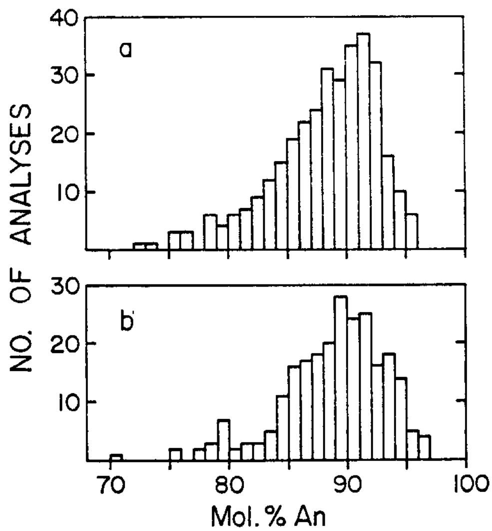

Fig. 3. (a) Distribution of compositions for random spot analyses in plagioclase of basalt fragments (minimum of 50 individual crystals). (b) Ditto for plagioclase in light matrix material.

to separate small grains in the mesostasis. The ilmenite grains are not compositionally zoned, but the Mg concentration is variable (0–2·4 wt.% Mg) from grain to grain is basalts and light matrix. High Mg values are typical of the ilmenites included in pyroxene (Table 2, no. 5) while the lowest values are representative of mesostasis grains. This trend reflects the decreasing Mg/Fe ratio of the residual liquids during crystallization.

Spinels present in the basalt fragments are generally poorer in Ti and richer in Cr than the Apollo 11 and 12 basalt spinels. The spinels included in early silicate precipitates such as olivine or pigeonite are not zoned and tend to be rich in Cr and poorest in Ti. A complete analysis of an early spinel (included in olivine) is given in Table 2, no. 6. For easy comparison, the compositional variation of the spinels may be expressed in terms of the three spinel 'molecules', Chr = $(Fe, Mg, Mn)Cr_2O_4$, Sp = $(Fe, Mg, Mn)Al_2O_4$, Ulv = $(Fe, Mg, Mn)_2TiO_4$, and the ratio of Fe:Mg:Mn. For example, analysis no. 6 may be expressed as $Chr_{57}Sp_{35}Ulv_8$ (72:24:4), accounting for 99·53 atom per cent of the analyzed cations. Spinel grains not included in olivine or pigeonite were in contact with liquid for a longer period and their zoning is pronounced, showing a dominant increase in Ti from core to edge. A typical range of zoning for these late spinel grains is $Chr_{57}Sp_{39}Ulv_4$ (80:19:1) in the core to $Chr_{48}Sp_{25}Ulv_{27}$ (85:14:1) at the edge. Note that the typical core spinel is poorer in Ti (Ulv) than the Apollo 11 and 12 spinels. The largest concentration of Ti in the 14321 spinel (9·05 wt.% Ti) was found in a grain with the composition formula $Chr_{41}Sp_{20}Ulv_{39}$ (87:12:1). Spinel grains in the olivine fragments of the light matrix are very similar to those found in the basalt clasts.

The basalt clasts contain rounded blebs of iron metal up to 50 μm diameter in a variety of associations from inclusions in olivine and pyroxene to distinct grains in the basalt mesostasis. The Ni and Co contents are variable between the limits 0–17·0 wt.% Ni and 0·2–3·5 wt.% Co with no detectable covariance. Metallic iron blebs in the mesostasis or associated with troilite have low ($<$0·5 wt.%) Ni contents while those included in early olivine or pyroxene crystals are invariably rich in Ni ($>$10·0 wt.%). Metal grains are difficult to identify in the light matrix material but the few that have been analyzed follow the same compositional trends observed in the basalts. Troilite occurs in the mesostasis of the basalts or as irregular grains filling the interstices between larger silicate crystals. The troilite is essentially FeS with no significant content of minor elements. Anhedral grains of cristobalite have also been analyzed in some of the basalt fragments. Aluminum is the only minor element present in significant concentrations (Table 2, no. 7).

The fine grained products of the last stage solidification of the basalt liquid are predictably variable in bulk chemistry and mineralogy, but the following components are generally present. (1) Cryptocrystalline patches rich in Si, K, and Ba. An average analysis of six separate areas is given in Table 1, no. 7. Occasionally these areas can be partly resolved with the microprobe into an SiO_2 phase and a K, Ba, and Al rich phase (presumably alkali feldspar). (2) Rare zircons up to 5 μm in diameter. (3) Fluorapatite and whitlockite, sometimes occurring in single composite grains. The whitlockite is greatly enriched in REE relative to the apatite as can be seen from the analyses in Table 3, nos. 1 and 2. The chondrite normalized REE abundance pattern for the whitlockite is shown in Fig. 4. (4) Anhedral grains of sphene have been identified in association with ilmenite and troilite in the basalt mesostasis. They are rare and all the grains are less than 5 μm in diameter. Sphene has not previously been identified in lunar materials. An average of six spot analyses from the two largest grains is given in Table 3, no. 7. The analysis in Table 3 corresponds to the molecular formula $(Ca_{0.958}X_{0.009}Fe_{0.033})_{1.000}$ $(Ti_{0.968}Zr_{0.010}Fe_{0.009}Al_{0.013})_{1.000}$ $(Si_{0.998}Al_{0.022})_{1.020}$ O_5 where X = REE.

3. *Microbreccia components*

Under this general heading we include all of 14321 other than the discrete igneous (basaltic) clasts and the light colored matrix, i.e. all of 14321 which existed as a fragmental rock prior to the lithification of 14321 proper. These components can be seen on a macroscopic scale as dark, rounded clasts in Fig. 1 (DUNCAN *et al.*, 1975b), and are designated here as microbreccia 3. Even at this scale it is possible

Table 3. Sphene and phosphate minerals in 14321

	1	2	3	4	5	6	7
SiO_2	—	—	0·46	0·31	0·72	—	30·19
TiO_2	—	—	—	0·00	—	—	38·93
Al_2O_3	—	—	—	0·27	—	—	0·91
Cr_2O_3	—	—	—	0·00	—	—	—
FeO	—	—	0·52	1·60	1·01	—	1·51
MnO	—	—	—	0·09	—	—	—
MgO	—	—	0·22	3·25	3·19	—	—
CaO	54·79	40·06	54·44	41·86	41·53	—	27·03
Na_2O	—	—	—	0·31	—	—	—
K_2O	—	—	—	0·07	—	—	—
P_2O_5	41·07	41·61	40·42	41·87	42·29	—	—
ZrO_2	—	—	—	—	—	—	0·62
Y_2O_3	—	—	0·24	—	2·93	—	—
La_2O_3	0·22*	1·36*	0·02	1·05*	1·10	1·33*	0·04*
Ce_2O_3	0·49*	3·11*	0·12	2·39*	3·13	2·99*	0·21*
Pr_6O_{11}	—	—	0·00	—	0·23	—	—
Nd_2O_3	0·40*	2·07*	0·05	1·58*	1·80	1·78*	0·16*
Sm_2O_3	0·09*	0·54*	0·01	0·41*	0·53	0·49*	0·04*
Eu_2O_3	0·01*	0·12*	0·00*	0·08*	0·07*	0·11*	0·01*
Gd_2O_3	0·15*	0·84*	0·04	0·58*	0·81	0·82*	0·08*
Dy_2O_3	0·13*	0·78*	0·04	0·28*	0·43	0·49*	0·15*
Ho_2O_3	—	—	0·01*	—	0·31*	—	—
Er_2O_3	—	0·18*	0·00*	0·00*	0·06*	0·27*	0·10*
Tm_2O_3	—	—	0·00*	—	0·00*	—	—
Yb_2O_3	0·04*	0·08*	0·00*	0·00*	0·00*	0·21*	0·06*
Lu_2O_3	—	—	0·00*	—	0·00*	—	—
F	2·39*	0·04*	3·05*	—	0·06*	—	—
Total:	99·78	90·79	99·64	96·00	100·20	—	100·07

1. Apatite, mesostasis basalt clast. Cl also present. 2. Whitlockite, composite grain with apatite, basalt clast. 3. Apatite, crystal clast in microbreccia. Cl also present. 4. Whitlockite, crystal clast in microbreccia. 5. Whitlockite, micronorite lithic clast. 6. Whitlockite, rhyolite glass lithic clast. 7. Sphene, mesostasis basalt clast, major elements—av. 6 analyses, REE—av. 2 analyses.

* Analyses corrected only for background.

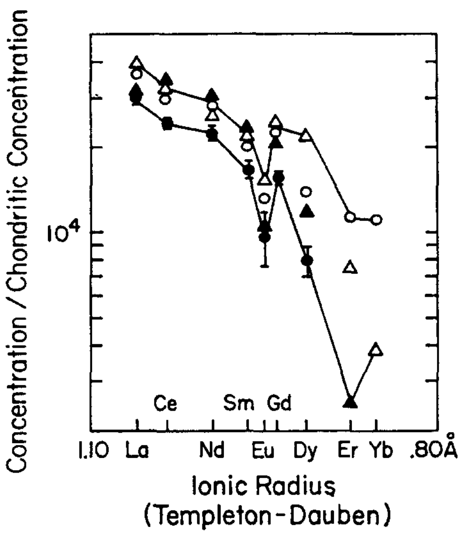

Fig. 4. REE abundances (chondrite normalized) in whitlockites of 14321. △, Basalt clast, no. 2; ▲, micronorite, no. 5; ○, glassy rhyolitic clast, no. 6; ●, crystal clast in microbreccia 3, no. 4. Numbers refer to analyses in Table 3.

to see smaller round inclusions incorporated within microbreccia 3. In thin section these are seen to be fragmental also, and we have designated them as microbreccia 2. Further study with the microscope and microprobe reveals that some of the microbreccia 2 clasts incorporate yet another fragmental rock type of noritic mineralogy that we have designated as microbreccia 1. The consistency in the order of inclusion among these three fragmental rock types as observed in seven thin sections points to a relative time sequence for their lithification which corresponds to the numbering sequence. We have also identified a single clast of a distinctive olivine microbreccia which cannot be placed in the same sequence except to note that it obviously predates microbreccia 3. Our study of the breccia components of 14321 points to at least three periods of non-igneous consolidation prior to the final lithification of 14321, and we will proceed to describe the petrography of each fragmental lithic unit under separate sub-headings starting with the oldest and proceeding in what we interpret to be chronological order.

3.1 *Microbreccia* 1. Small (1 mm maximum diameter) breccia fragments of distinctive 'noritic' mineralogy are found incorporated in microbreccia 2 (cf. CAMERON *et al.*, 1973). These light colored fragments are predominantly composed of plagioclase (An_{66-97}) and low-Ca pyroxene. Minor amounts of olivine (Fa_{14-21}), ilmenite, high-Ca pyroxene, Fe–Ni metal, and Ti chromite are also present. Figure 1e shows the textural characteristics of microbreccia 1 fragments. The larger crystals of plagioclase and low-Ca pyroxene are incorporated in a matrix of similar mineralogy. The fragments have been partly recrystallized during a period of intense thermal metamorphism, resulting in a matrix which is distinctly coarser than that of other breccias found in 14321. In some areas the matrix has been replaced by poikiloblastic pigeonite crystals as a result of the thermal metamorphism. The major element composition of microbreccia 1 is listed in Table 1, no. 4 and plotted

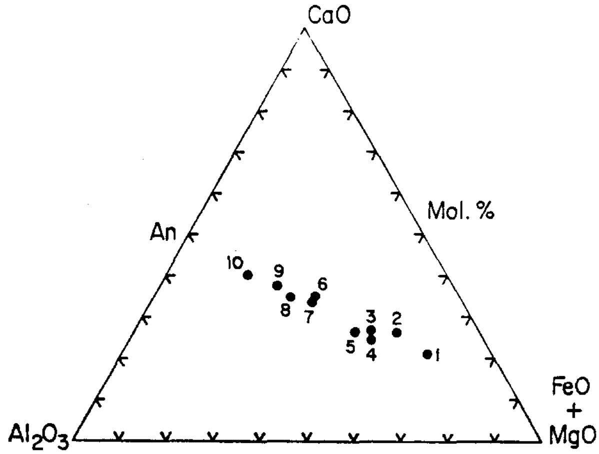

Fig. 5. Composition of various lunar rock types shown in ternary projection: 1, Apollo 14 mare basalts (Reid *et al.*, 1972); 2, 14321 basaltic clasts; 3, Fra Mauro basaltic glasses (Reid *et al.*, 1972); 4, microbreccia 2–3 dark matrix in 14321; 5, Apollo 12 gray mottled KREEP breccias (Meyer *et al.*, 1971); 6, microbreccia 1 in 14321; 7, highland basalts (Reid *et al.*, 1972); 8, micronorites in 14321; 9, feldspathic clast in 14321; 10, anorthosite (Reid *et al.*, 1972).

in Fig. 5. Microbreccia 1 is the oldest fragmental rock component of 14321 and was subjected to high grade thermal metamorphism before incorporation in microbreccia 2. Volumetrically the noritic microbreccia 1 clasts and the crystalline micronorite clasts of similar mineralogy (described below) represent the most abundant lithic component in microbreccia 3.

3.2 *Olivine microbreccia.* A 1·5 mm diameter fragment composed of olivine crystal clasts (Fa_{15-25}) in a matrix of similar olivine and minor Ti-chromite is enclosed in microbreccia 3. Its mineralogy and texture are very similar to that of fragment 14002,8 which was described as a dunite microbreccia by Taylor and Marvin (1971). The variability in olivine composition and the relatively large chromite grains of the matrix (20 μm) suggest an origin by fragmentation of more than one large olivine grain, perhaps of an earlier olivine-rich igneous rock.

3.3 *Microbreccias* 2 *and* 3. The larger microbreccia clasts in Fig. 1 (Duncan *et al.*, 1975b) appear to contain a number of rounded dark inclusions (microbreccia 2). In thin section (Fig. 1f) these dark inclusions are seen to be regions of the microbreccia composed primarily of a dark matrix, with relatively few of the larger crystal or lithic fragments that are incorporated in large numbers elsewhere. The matrices and included crystal and lithic fragments of the two breccia areas are not distinguishable on the basis of chemical composition and mineralogy, and the only significant difference seems to be the proportions of unresolvable fine grained dark matrix to larger fragments. The source of fragmental material was the same in both cases, but the darker inclusions formed first and were subsequently incorporated in similar material with larger average particle size. Accretionary structures have previously been recognized in Apollo 11 and 12 breccias (McKay and Morrison, 1971; Drake *et al.*, 1970). Alternating concentric bands of dark matrix and larger clastic fragments are characteristic of these structures (cf. Fig. 1f). The most straightforward interpretation of the relationship of microbreccias 2 and 3 is to consider that microbreccia 2 represents lunar accretionary lapilli formed in a base surge cloud and later incorporated within similar material during deposition of the cloud. This will be discussed further in the summary paper (DUNCAN *et al.*, 1975a). For present purposes we simply note that microbreccias 2 and 3 were formed sequentially but their textural relations and the similarity in mineralogy point to a nearly contemporaneous origin in a common rock-forming process.

3.3.1 *Dark matrix.* Fragments less than 15 μm in diameter are arbitrarily considered to be breccia matrix. Secondary electron scans of the matrix indicate that the lower size limit is below the resolution limit of the electron beam, i.e. $<0{\cdot}25$ μm. Defocused beam analyses have been taken of 14 separate areas of dark matrix in microbreccias 2 and 3. Approximately 15,000 μm^2 were covered in each area. We list the average of the 14 analyses and the standard deviation in Table 1, no. 3. The matrix is homogeneous in its major elements and there are no significant differences between the matrices of microbreccias 2 and 3. The average analysis is plotted in Fig. 5 and shown to be similar to the Fra Mauro basalt glass (REID *et al.*, 1972) and the Apollo 12 gray mottled KREEP breccias (MEYER *et al.*, 1971). The dark matrix composition is intermediate between mare basalt and the more feldspathic rock types such as norites, highland basalts, and anorthosites. Titanium is concentrated in small (<1 μm diameter) areas interstitial to the relatively larger fragments of the matrix. The grains are too small for quantitative analysis, but the Ti is probably in ilmenite which contributes to the dark color of the matrix. Potassium and phosphorus also appear to be concentrated in discrete areas of the matrix, but the individual phases are not resolvable. The dark breccia matrix is appreciably richer in K and P than the basaltic fragments of 14321.

3.3.2 *Non-clastic lithic fragments.* Sub-rounded, light colored clasts (maximum diameter 1·5 mm) of low-Ca pyroxene and plagioclase occur within microbreccias 2 and 3. These micronorites are usually granoblastic or hornfelsic with vestigial igneous sub-ophitic textures (Fig. 1g). The low-Ca pyroxenes show a total range in composition from $En_{87}Fs_{11}Wo_2$ to $En_{58}Fs_{38}Wo_4$ over all the micronorite fragments, but within individual fragments the range is very small and the individual crystals are not zoned. A representative analysis of a low Ca-pyroxene from a micronorite clast is given in Table 2, no. 14. Some of the low-Ca pyroxenes are poikiloblastic with inclusions of subhedral laths of plagioclase (An_{85-95}) which in turn contain small inclusions of low-Ca pyroxene. Minor phases include rare high-Ca pyroxene, ilmenite, Fe metal, zircon, and whitlockite. The latter two phases typically occur as poikiloblastic intergrowths with plagioclase crystals (Figs. 1h, i). Analyses are given in Table 3, no. 5 and Table 4, no. 2. The REE distribution of the whitlockite is very similar to that found for this mineral in other components of 14321 (cf. Fig. 4). We interpret the micronorite clasts as having been igneous prior to intense thermal metamorphism, in contrast to the originally clastic noritic microbreccia 1. The contrasting textures are clearly shown in Figs. 1e, g. The composition of the micronorite is listed in Table 1, no. 5 and plotted in Fig. 5. Along with the noritic microbreccia 1, the micronorite fragments predate microbreccia 2–3. It is quite possible that these two components of similar mineralogy originated in the same source rock, most probably a noritic basalt. The differences could very well simply be due to varying degrees of fragmentation of the source rock prior to thermal metamorphism. We have found one well-rounded noritic clast in microbreccia 3 which seems to confirm this interpretation. One portion of this clast is typical micronorite while the remainder has a fragmental texture identical to that of microbreccia 1.

Table 4. Zircons in 14321

	1	2
SiO_2	32·41	32·69
ZrO_2	65·40	66·85
Al_2O_3	0·15	0·20
FeO	0·47	0·09
MgO	—	0·02
CaO	0·07	0·15
P_2O_5	0·40	0·60
Total:	98·90	100·60
O=	4	4
Si	0·998	0·986
Zr	0·977	0·986
Al	0·005	0·011
Fe	0·012	0·002
Mg	—	0·001
Ca	0·002	0·005
P	0·010	0·015

1. Av. 3 analyses, zircon crystal clast in microbreccia. 2. Zircon, micronorite lithic clast.

No true anorthosites are present in the microbreccia, but a few plagioclase-rich fragments have been identified. Most of them consist of several large crystals of fractured plagioclase (maximum diameter 700 μm) with mosaic recrystallization textures (Fig. 1j). Small, anhedral olivines of composition Fa_{21-39} are also present. The plagioclase/olivine ratio is variable but plagioclase is always dominant. One additional feldspathic fragment has been analyzed. It consists primarily of plagioclase (An_{92-97}) with olivines (Fa_{29}) and low-Ca pyroxenes ($En_{73}Fs_{23}Wo_4$) of constant composition. Curiously, the olivine and pyroxene are very efficiently segregated. A defocused beam analysis of this fragment is listed in Table 1, no. 6 and plotted in Fig. 5 where, along with the micronorite and microbreccia 1, it can be seen to be significantly more 'anorthositic' than the dark matrix of microbreccias 2 and 3 and the 14321 basaltic fragments.

A variety of fragments have distinctly rhyolitic compositions. Many of these are devitrified brown glasses occurring as discrete rounded cryptocrystalline clasts, irregular grains, or veinlets cutting through plagioclase crystal fragments but not extending into the dark microbreccia matrix. The average composition of this component is listed in Table 1, no. 8. It is a low temperature melting system with approximately 90 per cent of the analysis expressible in terms of the granite components Or–Ab–Q. The textures indicate that it was liquid or at least in a plastic state, and that it was associated with some of the plagioclase crystal fragments before incorporation in microbreccia 3. The intimate association of molten or plastic (now glassy) rhyolitic material with plagioclase in lunar breccias has been noted before in connection with sample 12013 (DRAKE *et al.*, 1970). Some of these glassy fragments contain vesicles (Fig. 1k). In addition to silica and potassic feldspar, minor amounts of zircon and phosphate minerals have been identified in these devitrified glasses. The REE distribution in whitlockite from one of the rhyolitic fragments is given in Table 3, no. 6 and Fig. 4. One granoblastic 'micro-granite' clast occurs as the core of a fragment of microbreccia 2 (core of an accretionary lapilli?). A mosaic of equant grains of potassic feldspar and silica with minor zircon, apatite, and whitlockite make up this $1 \times 1{\cdot}5$ mm lithic fragment (Fig. 1l). A defocused beam analysis is given in Table 1, no. 9, and an analysis of the potassic feldspar is listed in Table 2, no. 15.

3.3.3 *Individual crystal fragments.* Large olivine clasts (up to 1·5 mm) are scattered throughout the dark matrix. They are fractured and rounded, and occasionally exhibit polysynthetic twinning due to shock. Typically their cores are homogeneously magnesian (Fa_{8-10}) and only the outer portions show normal zoning (up to Fa_{65}). Microprobe traverses across individual grains frequently reveal that the compositional zoning from edge to edge is strongly asymmetric, indicative of fragmentation and abrasion prior to incorporation in the microbreccia. The coarser olivine fragments are significantly larger than the olivine grains found in lunar olivine basalts, and their source rock was probably a coarser grained mafic to ultramafic igneous rock produced by slower cooling than was the case for the mare type basalts sampled to date.

Crystal fragments of pyroxene are abundant in the dark matrix, ranging from somewhat larger than 1 mm diameter down to matrix size. The majority of the analyzed grains are pigeonites and orthopyroxenes (Fig. 2b). All the larger grains ($>0{\cdot}5$ mm) are enstatite or bronzite. The relatively scarce Ca-rich pyroxenes exhibit exsolution lamellae of orthopyroxene and reaction rims of pigeonite (Fig. 1m). Analyses of a diopside host and its bronzite exsolution lamellae are given in Table 2, nos. 8 and 9. The reaction rims are pigeonite and are relatively constant in composition regardless of the composition of the core (Fig. 2c and Table 2, nos. 10 and 11). Wherever the core is twinned, the pigeonite reaction rim also exhibits dual extinction under crossed polarizers. Evidently the core crystal exerted epitaxial control on the new pyroxene crystal growth during reaction with the surrounding matrix material. The exsolution lamellae and the reaction rims are indicative of a period of thermal metamorphism for microbreccia 3 (WARNER, 1972). For reasons to be discussed in DUNCAN *et al.* (1975a), this metamorphic episode is considered to be less intense than the earlier one imposed on microbreccia 1.

Plagioclase fragments are very abundant. The total range of An content is 51–98 mole % with an average value of An_{88} (cf. Fig. 3, showing the range of plagioclase composition in the basalt clasts and light matrix). The plagioclase clasts in microbreccia 3 can also be distinguished from the basalt plagioclase by their lower content of Fe and Mg (Table 2, no. 12). The close approach to ideal feldspar stoichiometry [(Si–Na–K)-2 = 0·0122 and 1-(Al–Ca–Fe–Mg) = 0·015 based on 8 oxygens for analysis no. 12 which is typical] in thermally metamorphosed microbreccia, as compared to the significant departures from ideal stoichiometry in plagioclase from mare-type basalts is in agreement with previous findings (DRAKE *et al.*, 1970). Thin reaction rims are visible under cathodoluminescence on many of the plagioclase fragments. In one example where the rim was wide enough for probe analysis it had a composition of An_{90} in contrast to the core of An_{75}. Evidently thermal metamorphism of microbreccia 3 caused partial equilibration with the matrix resulting in more calcic plagioclase. Offsetting of twin lamellae by microfaulting is common. Some fragments have twin lamellae similar to those described in CHAO *et al.* (1970) as mechanical twins. Less common shock induced lamellar structures as illustrated in SHORT (1970) are also present. Devitrified glass fragments of plagioclase composition are relatively common. One such fragment is illustrated in Fig. 1n. These probably represent impact glasses which have devitrified during thermal metamorphism. Some fractures in the plagioclase clasts are filled with dark microbreccia matrix, indicating that some fracturing occurred during the lithifaction of microbreccia 3.

The ilmenite clasts of microbreccia 3 attain a maximum dimension of 1·3 mm, significantly coarser than the basalt ilmenites. Chemically they are distinguished by somewhat higher Mg content (1·0–3·0 wt. % Mg). Two spinel types are found as rare crystal fragments. Ti-chromites are similar in composition to those found in the basalt clasts. A few pleonaste spinel fragments are conspicuous in the microbreccia because of their pink to dark red color in thin section. A complete analysis of one of these grains is given in Table 2, no. 13. In terms of the three spinel 'molecules' defined previously this analysis may be expressed as $Chr_{9\cdot9}Sp_{88\cdot9}Ulv_{1\cdot2}$ (27:73:0), accounting for 98·3 per cent of the cations analyzed. These spinels are of distinctly different composition than those previously found in typical lunar mare basalts and very probably have a different origin.

Metal occurs as discrete grains, as inclusions in olivine clasts, and attached to small troilite fragments. There is a large variation in Ni content (1·35–42·4 wt. %) which is not systematically related to the mode of occurrence, but is positively correlated to the concentration of Co (0·22–2·3 wt. %) as seen in Fig. 6. We have included in Fig. 6 the 95 per cent confidence band for the Ni–Co correlation in meteoritic metal as discussed in GOLDSTEIN and YAKOWITZ (1971), and according to this criterion only a small proportion of the metal grains could be of meteoritic origin. Two of the analyzed grains, both included in Mg-rich olivine clasts, are extremely rich in Ni and Co and plot off scale. The analyses show a bimodal frequency distribution for Ni with broad maxima at 6–9 and 15–18 wt. %. With the exception of four composite grains, all the particles are homogeneous. A typical example of a composite metallic grain is illustrated in Fig. 1o, and the composition tie lines are given in Fig. 6. There is no measurable compositional gradient within the Ni-poor and Ni-rich portions, and the boundary is very sharp (in all cases the Ni contrast indicated by the tie-lines is fully developed over 2–3 μm). The polished surfaces of these composite grains show no contrasts in reflected light microscopy, but a 5-sec etch in dilute nitric acid is sufficient to reveal the phase boundary. The structural states of the two phases are not known, and the compositional data are not uniformly compatible with kamacite–taenite equilibration at sub-solidus temperatures. The textural relations are not particularly suggestive of solid-state

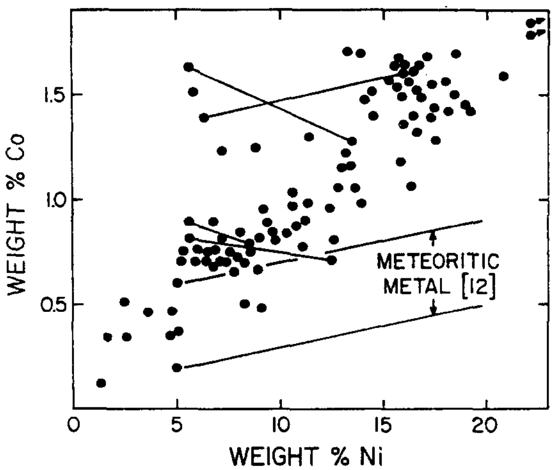

Fig. 6. Nickel and cobalt concentrations in metallic particles of microbreccia 2–3. Concentrations expected in meteoritic metal according to criteria discussed in GOLDSTEIN and YAKOWITZ (1971) are also shown. Tie-lines join coexisting compositions in composite grains. Note that two of the analyses plot off scale (2·1% Co, 38·7% Ni; 2·2% Co, 42·4% Ni).

exsolution, and it is equally likely that the compositions of the composite grains may be partly inherited from a rapidly cooled liquid–solid system.

Anhedral grains of troilite with or without associated metal form a very minor component of microbreccias 2 and 3. Apatite, whitlockite, zircon (Table 4, no. 1), and K-feldspar have also been identified. No silica fragments were observed. These minor constituents are generally fine grained, but at least one rounded and heavily fractured apatite grain of 500 μm was observed. The apatites are chemically similar to those found in the basalts, poor in REE but containing appreciable fluorine (Table 3, no. 3). A typical breccia whitlockite analysis is given in Table 3 no. 4 and its REE distribution pattern is shown in Fig. 4. The whitlockite analyses shown in Fig. 4 from four distinct lithologies (micronorite, basalt, microgranite, and dark microbreccia) all show the same REE distribution pattern, with strong enrichment of the light REE and a small but consistent negative Eu anomaly.

SUMMARY

Lunar sample 14321 is a large (9·0 kg, nicknamed 'Big Bertha' by the Apollo 14 crew) heterogeneous rock containing many chemically, mineralogically, texturally, and chronologically distinct components. We have classified the components according to the above criteria after a petrographic examination of seven thin sections. This classification is only as complete as the seven thin sections are representative of the bulk sample.

Three major components have been brought together in the last stage formation of 14321: igneous basaltic fragments, clasts of an older breccia, and a lighter colored matrix binding the other two components together to form the present-day bulk 14321 polymict breccia. The basaltic fragments have similar bulk compositions but display a variety of textures ranging from fine grained ophitic (holocrystalline) to vitrophyric. The textural variation and compositional similarity suggest that these clasts represent a sampling from different locations (with different cooling rates) within a single lava cooling unit or a genetically related set of cooling units. The microbreccia clasts incorporate at least two older sets of fragmental clasts. The oldest of these (microbreccia 1) is noritic, consisting primarily of orthopyroxene and plagioclase, and has been through a period of intense thermal metamorphism. Within the microbreccia clasts areas of very fine grained dark matrix are outlined by concentric bands of lighter colored silicate crystal fragments. Such areas are usually cored by large crystal fragments or lithic fragments (e.g. microbreccia 1). These areas have been designated as microbreccia 2 and interpreted as lunar accretionary lapilli structures. In the mineralogy of the crystal fragments and in the bulk chemical composition of the dark matrix, microbreccia 2 is identical to the surrounding breccia material which we designate as microbreccia 3. The two units were formed in a common impact fragmentation process and are nearly contemporaneous, microbreccia 2 lapilli structures having been incorporated in microbreccia 3 during the lithification of the debris blanket created by the impact event. Microbreccia 2–3 components have also been thermally metamorphosed but not as intensely as microbreccia 1. The bulk microbreccia clasts (microbreccia 3) are well rounded and have been subjected to an efficient abrasion process subsequent to fragmentation.

The light colored matrix which surrounds and separates the basaltic and microbreccia 3 clasts is composed of crystal and lithic fragments. Most of the crystal clasts large enough for microprobe analysis are identical in composition to the corresponding phases in the crystalline basalt clasts. Most of the recognizable lithic fragments of the light matrix are basaltic, similar in textural, mineralogical, and compositional detail to the basalt clasts. Much of the light matrix material of 14321 has been derived from the fragmentation of the lava cooling unit(s) which gave rise to the distinct basalt clasts of 14321. The remainder of the light matrix is composed of fragments of microbreccia 3 (Duncan *et al.*, 1975b). The crystals of the light matrix and the basalt clasts are only weakly shocked, and this last fragmentation process seems to have been a relatively mild event which was not followed by thermal effects.

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